

CSE499A – Section 15 – Group 03

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CityPulse: Intelligent Parking & Traffic Control Platform

Weekly Progress Report – Update 5 (Sprint 5 in progress)

I. SPRINT OVERVIEW

In the last week we advanced three main areas: AI research, backend API readiness, and frontend owner-portal capability. AI work added the VPS-Net occupancy training output and a formal review of the VPS-Net parking-slot detection paper. Backend progress included Swagger DTO documentation and authentication/parking-space DTO refinements, while the frontend gained owner dashboard, camera, revenue, and settings pages plus new YOLOv8 PKLot reproduction support.

II. AI RESEARCH AND MODEL VALIDATION

The AI team completed the first integration of the VPS-Net occupancy classification workflow into the repository. This includes training output, notebook analysis, and a paper review for the VPS-Net approach to warped slot classification. We also added Colab reproduction materials for YOLOv8 on the PKLot dataset and started documenting the YOLOv8 parking detection pipeline in detail.

- Added ‘applications/ai/ai-models/parking-slot-detection/VPS-Net/notebooks/vps-net.ipynb’ with training and validation diagnostics.
- Added ‘others/paper-reviews/VPS-Net-review.pdf’ to capture literature review findings and future research directions.
- Added YOLOv8 PKLot reproduction notebooks and configuration files for comparative model development.
- Notebook training summary: customized DCNN reached a best validation accuracy of **98.53%** after 323 epochs (global step 90117); reproduced model inference time **0.60 ms** and model size **9.01 MB**. Training curves and CSV logs are available under the VPS-Net notebook outputs.
- Paper review highlights: VPS-Net’s detection + perspective transform pipeline and a lightweight customized DCNN achieved reported accuracy of **99.48%** with a model size of **9.4 MB** and inference time **0.81 ms** (authors’ evaluation); our reproduced model reached **98.53%** val accuracy with **0.60 ms** inference and **9.01 MB** size on the current split.
- Notes: the reproduced model shows strong performance but remains slightly below the paper’s reported occupancy-classification accuracy; next steps are fine-tuning on local footage and evaluating runtime on target hardware.

III. BACKEND API PROGRESS

Backend work focused on API contract quality and authentication flows. Swagger documentation now covers the main DTOs used in auth and space management, improving API discoverability for frontend integration and QA. We also updated package dependencies and corrected the production entry point to ensure deployment readiness.

- Integrated Swagger documentation for data transfer objects (DTOs) in ‘applications/web/backend’.
- Updated auth data transfer objects for login, register, refresh, forgot-password, and reset-password.
- Refined parking space data transfer objects for create and update operations.
- Ensured the backend package manifest and production entry point are consistent with deployment build steps.

IV. FRONTEND AND USER EXPERIENCE

The frontend team implemented the owner portal and documentation support for the next phase of the product. New owner-facing pages provide camera management, revenue history, and facility settings. Supporting UI screenshots and a new React dependency for toggle controls were also added.

- Added owner dashboard page and routing.
- Added owner-facing pages for cameras, revenue, and settings.
- Added frontend screenshot documentation for Week 8.
- Added ‘@radix-ui/react-switch’ dependency to support modern toggles.

V. DOCUMENTATION AND REPRODUCIBILITY

We improved documentation coverage across AI and frontend research efforts. The repository now includes explicit reproduction notes for the YOLOv8 PKLot experiments and a curated paper notes document for decentralized parking lot occupancy detection.

VI. PLAN FOR NEXT WEEK

- Complete RTSP ingestion worker design for the AI service.
- Finalize password reset and auth recovery endpoints in the backend.
- Continue local parking footage collection for Phase 3 fine-tuning.
- Integrate the new owner portal pages with live backend APIs.
- Perform a full reproduction of the VPS-Net slot-detection pipeline (YOLO-style slot detection), then integrate the reproduced slot detector with our custom DCNN for occupancy classification.

REFERENCES

- [1] R. R. Prova *et al.*, “A Real-Time Parking Space Occupancy Detection Using Deep Learning Model,” *IEEE IEMTRONICS*, 2022. DOI: [10.1109/IEMTRONICS55184.2022.9795771](https://doi.org/10.1109/IEMTRONICS55184.2022.9795771)
- [2] G. Amato *et al.*, “Deep learning for decentralized parking lot occupancy detection,” *Expert Systems with Applications*, vol. 72, 2017.
- [3] B. Yuldashev *et al.*, “Parking Lot Occupancy Detection with Improved MobileNetV3,” *Sensors*, vol. 23, no. 17, 2023. DOI: [10.3390/s23177642](https://doi.org/10.3390/s23177642)